intro001 — shot list

Every shot in the assembly, in order. **60 shots, 12:22.13**, 23.976p, 1920×1080.

Generated from `picks.json`, so it cannot drift from the timeline. *Sound* is where the audio comes from; *picture* is only filled in where the shot is a split edit and the picture comes from somewhere else.

Sources

Clip	Shots	On screen
`MVI_0019.MOV`	28	6:05.23
`MVI_0021.MOV`	28	6:02.19
`MVI_0016.MOV`	4	0:13.19

Music

Track	In	Out	Level
Audiio_AsTallAsPine_ATAP_TwinLakes	0:00.00	0:31.23	16 dB under the voice
Audiio_SK_Commercial_TwoBigReasons30	11:50.09	12:22.13	16 dB under the voice

Shots

#	At	Dur	Sound	Picture	What it is
1	0:00.00	2.7s	`MVI_0016.MOV @ 28.86`	**starter-jar-rise_prepped.mov @ 0.00**	hook — yeast bubbling in action (take 4) · sound +19.7 dB
2	0:02.17	2.9s	`MVI_0016.MOV @ 66.36`	**MVI_0016.MOV @ 34.28**	hook — way too excited about yeast (take 3) · sound +19.7 dB
3	0:05.15	4.2s	`MVI_0016.MOV @ 92.55`	—	name + promise (take 5) · sound +19.7 dB
4	0:09.19	4.0s	`MVI_0016.MOV @ 112.28`	—	as humans, thousands of years · sound +19.7 dB
5	0:13.19	8.0s	`MVI_0019.MOV @ 40.21`	—	records are murky
6	0:21.18	3.3s	`MVI_0019.MOV @ 50.97`	—	about 3,000 years ago / 1,000 BC

#	At	Dur	Sound	Picture	What it is
7	0:25.00	2.1s	`MVI_0019.MOV @ 67.15`	—	Egypt, buried with pharaohs
8	0:27.02	6.9s	`MVI_0019.MOV @ 78.37`	—	jars, yeast on the side, DNA
9	0:34.00	14.8s	`MVI_0019.MOV @ 85.54`	—	Asia into Europe, no way to know which was first
10	0:48.20	6.4s	`MVI_0019.MOV @ 101.94`	—	DNA is mixed in with everything else
11	0:55.05	27.0s	`MVI_0019.MOV @ 120.83`	—	passed brewer to baker, off grapes, in the air
12	1:22.05	15.0s	`MVI_0019.MOV @ 153.03`	—	successful batch passed on, dud batch dropped
13	1:37.05	15.8s	`MVI_0019.MOV @ 170.38`	—	that is selective breeding
14	1:53.00	26.4s	`MVI_0019.MOV @ 186.56`	—	spontaneous generation → Pasteur's gooseneck flask
15	2:19.08	19.9s	`MVI_0019.MOV @ 228.35`	—	Meyen, 1838 → Saccharomyces cerevisiae
16	2:39.05	23.6s	`MVI_0019.MOV @ 251.33`	—	the jars: mead, instant, active dry, fresh, pre-ferment
17	3:02.20	6.0s	`MVI_0019.MOV @ 279.70`	—	~90% of the yeast we use is this one bug
18	3:08.19	9.8s	`MVI_0019.MOV @ 286.08`	—	sourdough changes batch to batch, mother to mother
19	3:18.14	9.2s	`MVI_0019.MOV @ 301.01`	—	other yeasts + bacteria are what make sourdough different
20	3:27.18	15.1s	`MVI_0019.MOV @ 310.39`	—	what yeast is — fungus, single cell, eukaryotic
21	3:42.21	19.4s	`MVI_0019.MOV @ 331.66`	—	budding — mother, daughter, mitosis
22	4:02.06	21.5s	`MVI_0019.MOV @ 352.52`	—	same guy, home bakery to the vineyards
23	4:23.18	13.4s	`MVI_0019.MOV @ 398.44`	—	it thrives where it is fed — starches and sugars

#	At	Dur	Sound	Picture	What it is
24	4:37.03	8.1s	`MVI_0019.MOV @ 412.41`	—	54 °F to just over 100 °F
25	4:45.06	1.8s	`MVI_0019.MOV @ 420.92`	—	underneath 54 degrees —
26	4:47.00	12.9s	`MVI_0019.MOV @ 428.76`	—	— it hibernates until it warms back up
27	4:59.21	13.0s	`MVI_0019.MOV @ 443.28`	—	over 100 °F, and 140 °F kills it
28	5:12.20	11.4s	`MVI_0019.MOV @ 458.62`	—	oven spring, and no live yeast in baked bread
29	5:24.05	33.2s	`MVI_0019.MOV @ 620.49`	—	CHEMISTRY — starch → amylase → sucrose → invertase
30	5:57.08	7.0s	`MVI_0019.MOV @ 670.04`	—	CHEMISTRY — zymase → CO_2 + alcohol
31	6:04.09	9.6s	`MVI_0019.MOV @ 697.91`	—	both burn off when you bake it
32	6:14.00	5.8s	`MVI_0019.MOV @ 713.25`	—	the CO_2 is why champagne is bubbly
33	6:19.18	3.7s	`MVI_0021.MOV @ 39.25`	—	fat, salt and spice in the dough —
34	6:23.10	12.7s	`MVI_0021.MOV @ 49.80`	—	— get between the yeast and its food
35	6:36.03	18.7s	`MVI_0021.MOV @ 62.52`	—	so factor in more proofing time
36	6:54.20	5.2s	`MVI_0021.MOV @ 87.84`	—	a pre-ferment is flour, water, yeast
37	7:00.00	13.7s	`MVI_0021.MOV @ 93.01`	—	you do not need sugar to activate it
38	7:13.17	17.1s	`MVI_0021.MOV @ 107.15`	—	sugar buys speed and costs you a sweeter bread
39	7:30.19	12.1s	`MVI_0021.MOV @ 136.76`	—	if it is all one organism, why so many products?
40	7:42.20	1.8s	`MVI_0021.MOV @ 149.02`	—	it needs access to food

#	At	Dur	Sound	Picture	What it is
41	7:44.15	4.8s	`MVI_0021.MOV @ 158.53`	—	it comes down to how often you need it
42	7:49.11	20.1s	`MVI_0021.MOV @ 163.41`	—	a working baker uses it daily → fresh yeast
43	8:09.14	33.9s	`MVI_0021.MOV @ 184.02`	—	an hour from a grocery store → active dry / instant
44	8:43.11	1.5s	`MVI_0021.MOV @ 217.93`	—	and those shops —
45	8:44.22	5.3s	`MVI_0021.MOV @ 221.39`	—	— probably do not sell much yeast
46	8:50.05	17.8s	`MVI_0021.MOV @ 232.27`	—	stock that dies on the shelf is not viable
47	9:07.23	22.4s	`MVI_0021.MOV @ 250.42`	—	brewer's yeast — marketing, convenience, the customer
48	9:30.09	19.1s	`MVI_0021.MOV @ 272.98`	—	if you want the best loaf, instant vs active dry differs
49	9:49.10	11.7s	`MVI_0021.MOV @ 309.56`	—	CHEMISTRY — drying damages it; it releases glutathione
50	10:01.03	5.9s	`MVI_0021.MOV @ 329.08`	—	CHEMISTRY — glutathione attacks the gluten matrix
51	10:07.00	16.5s	`MVI_0021.MOV @ 335.13`	—	so the bread is not as stretchy
52	10:23.12	8.4s	`MVI_0021.MOV @ 358.32`	—	and it dries out faster
53	10:31.22	6.5s	`MVI_0021.MOV @ 367.16`	—	day to day, does it matter? probably not
54	10:38.09	17.1s	`MVI_0021.MOV @ 379.80`	—	but if it does, try switching to fresh
55	10:55.10	18.2s	`MVI_0021.MOV @ 398.52`	—	otherwise use what you can get
56	11:13.14	6.5s	`MVI_0021.MOV @ 416.87`	—	it is all the same — look at sourdough
57	11:20.01	14.1s	`MVI_0021.MOV @ 449.95`	—	sourdough was what was available out west

#	At	Dur	Sound	Picture	What it is
58	11:34.03	21.4s	`MVI_0021.MOV @ 464.05`	—	you can start a sourdough from active dry
59	11:55.11	21.0s	`MVI_0021.MOV @ 486.28`	—	close — it feels like a mystery, it is one bug
60	12:16.11	6.1s	`MVI_0021.MOV @ 514.89`	—	close — CO_2 and ethanol, a happy life

What each shot says

1. 0:00.00 — This is yeast bubbling in action.

2. 0:02.17 — This is me, way too excited about yeast.

3. 0:05.15 — Katie, I'm a food scientist, and this is my deep dive into yeast.

4. 0:09.19 — As humans, we started using yeast thousands of years ago.

5. 0:13.19 — The records for when yeast first showed up is kind of murky from what I could find in my research.

6. 0:21.18 — about 3,000 years ago, about 1,000 BC.

7. 0:25.00 — we see some in Egypt buried with pharaohs.

8. 0:27.02 — They have jars, yeast on the side from what the DNA test can solve.

9. 0:34.00 — We also have yeast showing up in the archaeological records from Asia into Europe, all over the place. We don't have exact ways of knowing which one was first, where they were, if we got all of the records, but we know it's all over the place.

10. 0:48.20 — It's hard to pull out the DNA because it's also mixed in with a bunch of other things from my understanding.

11. 0:55.05 — Then we have this understanding of yeast. It's being passed between brewers and bakers. It's being collected from the air. It's being collected from outside of grapes and introduced into wine. It's moving around. It's passed from person to person, baker to brewer, brewer to baker. It's in meat. It's in wine. It's in beer. It's in our sourdough starters. It's just all around.

12. 1:22.05 — what they're doing is they're passing it from successful batch passed on to someone else's and they make a successful batch and if it's a dud batch they don't pass it on because who wants to do anything with a dud batch

13. 1:37.05 — And so what we're getting is selective breeding of good yeast, yeast that ferments, that makes the bread bubble, makes the alcohol happen, is being passed around. And that is what we've used for years and years and years.

14. 1:53.00 — and it was just thought it was spontaneous generation that's what it was until we get to louis pasteur louis pasteur comes in and he does his gooseneck bottle experiment and proves that it's not

spontaneous generation that is making food come alive it is bacteria it's fungi it's a bunch of microscopic little things in our food and that's it that's all we know for a while with about yeast

15. 2:19.08 — And then we get Franz Julius Ferdinand Mayen. in 1838. He comes in and he discovers what we know for most of our yeast today, Saccharomyces cerevisiae. And that is the yeast that we see in our active dry yeast. It's what we find in our fresh yeast, our instant yeast, and most of our sourdoughs.

16. 2:39.05 — I have marketing for yeast for mead. It's Saccharomyces cerevisiae. I have instant dry yeast. It's Saccharomyces cerevisiae. I have active dry yeast with a little sugar in it. Guess what? Saccharomyces cerevisiae. Fresh yeast. Saccharomyces cerevisiae. I even have what could be the start of a sourdough starter. Here, it's a pre-ferment. It is yeast, Saccharomyces cerevisiae mixed with flour and water.

17. 3:02.20 — It is like 90% of all the yeast that we use is this one bug.

18. 3:08.19 — And in sourdoughs, we have, it's in most of the sourdoughs, but sourdough is where we see it change from batch to batch, mother to mother.

19. 3:18.14 — there might be some other yeast inside of those batches and they have other bacteria and that's what makes sourdough just a little bit different than your everyday yeast, flour, pre-ferment like I have here.

20. 3:27.18 — Okay, so what is yeast in and of itself? Yeast in and of itself is a, it's a small little bug, it's a fungi, it's single cell organism, it's eukaryotic,

21. 3:42.21 — It reproduces itself by budding where a mother... cell will start to grow a little bud off of it for a daughter cell and the daughter cell once it gets big enough mitosis happens and it splits and then it and then it continues to grow and those split and those split

22. 4:02.06 — and that is how Saccharomyces cerevisiae splits and grows and reproduces all over we use it everywhere from our home bakeries to the brewers down the road to the vineyards in California and France, everywhere, Saccharomyces cerevisiae, same guy, again and again and again.

23. 4:23.18 — So it will bud and reproduce in an environment that it thrives in. It likes it to be nutritionally dense for it. That's where it likes having starches and sugars to break down and give the energy that it needs to continue to grow.

24. 4:37.03 — It likes its temperature between 54 degrees Fahrenheit and just over 100 degrees Fahrenheit.

25. 4:45.06 — Underneath 54 degrees,

26. 4:47.00 — it goes back into a little shell of itself and hibernates and it just tucks away and takes a nap until it warms back up and is a good environment for it to grow and thrive and that's how it survives not being able to move from environment to environment.

27. 4:59.21 — Over 100 degrees it can still grow until it gets to about 140 degrees Fahrenheit and that's when baking happens and it kills it off in the oven

28. 5:12.20 — You might get oven spring from the yeast still growing and other things happening but then it stops and you're not eating fresh yeast when you eat bread because you've killed it off.

29. 5:24.05 — Okay, so you have your starch and it breaks down with alpha amylase and beta amylase. And we'll get into that in another video. And then that breaks down to sucrose. And sucrose... exists around

the yeast and the yeast itself provides invertase. And invertase is an enzyme. If it ends in "-ase", it's an enzyme. And so the invertase goes in to sucrose and breaks it down into glucose and fructose.

30. 5:57.08 — You get zymase and it breaks it down into your CO2 and your alcohol. That's your carbon dioxide and your alcohol.

31. 6:04.09 — And then when you bake the bread, the alcohol and the CO2 kind of burn off and leave the bread. So they're not behind left in the bread after you've cooked it.

32. 6:14.00 — And you also have your CO2. That's where you get your bubbly wines like your champagne.

33. 6:19.18 — In your dough, you have fats, salts, spices.

34. 6:23.10 — in the way between the yeast and the sugar and the starches that it wants to eat. And so it slows it down. It takes longer. It's harder for the yeast to reach its food source. It can get there. It just takes more time.

35. 6:36.03 — So when you're cooking with added fat, salt, spices to a a dough you just need to factor in that it's going to take more time for it to proof for it to get to what it needs because it needs starches sugars it needs water that's all it needs

36. 6:54.20 — A pre-ferment is flour, water, yeast. That's all is needed.

37. 7:00.00 — I know some yeast says it needs sugar to activate active dry yeast. They recommend that. Really, if you give it time, you don't need the sugar in it. You don't have to add the sugar to your dough unless that is a, you want a sweeter dough.

38. 7:13.17 — So it just, how much time do you want? If you need to bake dough and you want it to rise a little bit faster, you can add sugar to it. It's just going to be a sweeter bread, but it will rise a little faster, get going a little faster, shorten your proofing time.

39. 7:30.19 — Saccharomyces cerevisiae, I said it's all the same thing, but why? Why do we have so many different forms when it's all the same guy? Well, he likes to go dormant. He might die.

40. 7:42.20 — He needs access to food sources.

41. 7:44.15 — it comes down to how frequently does the person using it need to have access to the yeast.

42. 7:49.11 — A baker is using yeast day in and day out. They can have fresh yeast on hand because they might not want to wait for a dry yeast to come building back up into active yeast that is moving and flowing. Having fresh yeast will shorten their cook time, their baking time, their proof time, and it makes it easier.

43. 8:09.14 — If you have a baker in Kansas that lives on a farm that takes an hour to get to their local grocery store, having fresh yeast... is not practical because they need something that can stay on hand, that lasts longer. That's where you get your active dry and your instant yeast needs. It's a casual baker, a person that doesn't want something that is gonna go and die within a couple weeks, a week. They need something that lasts longer because of the effort it takes to get to the grocery store.

44. 8:43.11 — Also, those grocery stores

45. 8:44.22 — probably don't sell a lot of yeast because they might not have as frequent customers.

46. 8:50.05 — And so having fresh yeast on hand and having it die on hand again and again might not be financially viable. So they have the dry yeast on hand so that way when the customer comes in, it's available, it's not dead, it's... It's what those people need.

47. 9:07.23 — But if you have yeast on hand and you have someone that's wanting to learn how to brew, they're searching, how do I find yeast? And they get brewer's yeast. Yes, I'm brewing. Let me buy brewer's yeast. It's marketing sometimes. Sometimes it's convenience. Sometimes it's who the customer is. But if you know what you're looking for, you can buy what you can find and work with it.

48. 9:30.09 — If you're really wanting to deep dive into making the best, most perfect bread, the most premium bread, you might care what type of yeast, because there is a little bit of a difference with instant and active dry.

49. 9:49.10 — And so the yeast might become damaged from sitting and being dried out and when it comes up and back to life, it can release something called glutathione.

50. 10:01.03 — And what that can do is it can get into the gluten matrix and attack it a little bit.

51. 10:07.00 — So you have your your instant dry yeast, what you might find is one of the, it's available, it's findable, but one of the offsets might be that the bread is not as stretchy.

52. 10:23.12 — The bread might dry out a little faster because the starch has a harder time holding onto water with the GSH involved.

53. 10:31.22 — And so day to day, does it really matter? Probably not with the bread that you want to make.

54. 10:38.09 — But if it really matters, if you're finding that your bread is drying out, it's not as stretchy, it's not as elastic as you want the bite to be, try seeing if you can find fresh yeast and switching to that. That might help fix the problem that you run into.

55. 10:55.10 — But if you're happy with your bread, it doesn't matter if you have access to fresh yeast. Active dry works great. If you have fresh yeast available but you can't find for some reason the active dry or instant yeast, go ahead and use that. Use what's available.

56. 11:13.14 — It's all the same. It's what we've done for hundreds of years. I mean, look at our ancestors who made sourdough.

57. 11:20.01 — Sourdough was what was available. They didn't have to ship back, ship yeast from back east to the hard to reach people. And so that's why some people, sourdough is even better than active dry yeast for them.

58. 11:34.03 — And if you're struggling because you want a fresh yeast, and all you can find is active dry, try just making sourdough from your active dry. Get it going, get it started, let the yeast multiply, be happy for a few days, and then you have a little pre-ferment already made and go from there.

59. 11:55.11 — It can be played with. Yeast is not something that has to be a hard and fast rule. It feels like a mystery, but really all it is is the same little Saccharomyces cerevisiae bug again and again and again and again. He's happy. All the little bug wants is to go and eat sugars and fart.

60. 12:16.11 — produce carbon dioxide and ethanol. It just wants to live a happy life and it's not all that complicated.